

Raphaël GASTALDELLO

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Research Interests

My research interests lie in the mathematical analysis of problems arising in statistical physics, at the interface of analysis, probability, and numerical methods. I am particularly interested in the efficient computation of transport coefficients and rare event probabilities, notably using advanced sampling techniques such as the Adaptive Multilevel Splitting algorithm. My work also concerns interacting particle systems and their mean-field limits, as well as the study of convergence to equilibrium using hypocoercive techniques.

Research Experience

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| 2023–2026 | Ph.D. in Mathematics , <i>CNRS, CERMICS (ENPC), INRIA team MATHE-RIALS, France.</i>
"Efficient computation of stochastics dynamics out of equilibrium"
Supervised by G. Stoltz and U. Vaes. Funded by collaborative project CNRS / Imperial College London. Collaboration with G.A. Pavliotis.
Topics: modeling, mathematical physics, interacting particle system, transport coefficients, PDE analysis (hypocoercivity), numerical simulations (Julia). |
| 2023 (5 months) | Master thesis , <i>CERMICS (ENPC), INRIA team MATHE-RIALS, France.</i>
"Computation of transport coefficients using Green–Kubo formula using Gir-sanov’s reweighting"
Supervised by G. Stoltz and U. Vaes.
Topics: modeling, transport coefficients, importance sampling, Numerical sim-ulations (Julia). |
| 2022 (2 months) | Research internship , <i>Institut Elie Cartan de Lorraine (IECL), Champs-sur-Marne, France.</i>
"Blaschke-Santaló diagram of area, perimeter and moment of inertia for for convex domains"
Supervised by A. Henrot.
Topics: Calculus of variations, convex analysis, numerical simulations (Matlab). |

Education

MSc in applied Mathematics

Sorbonne Université, Paris, France, 2022 –2023

BSc and MSc in Mathematical Engineering

Ecole Nationale Supérieure des Mines de Nancy, Nancy, France, 2020 – 2023

Publications

- [1] N. Blassel, L. Carillo, S. Darshan, R. Gastaldello, A. Iacobucci, E. Marini, R. Santet, X. Shang, G. Stoltz, and U. Vaes. “Mathematical analysis and numerical methods for the computation of transport coefficients in molecular dynamics”. In: *arXiv preprint arXiv:2605.10507* (2026).
- [2] R. Gastaldello, G. A. Pavliotis, G. Stoltz, and U. Vaes. “On the diffusive-mean field limit of a kinetic weakly interacting particle system”. In: *arXiv preprint arXiv:2607.15896* (2026).
- [3] R. Gastaldello, G. Stoltz, and U. Vaes. “Dynamical reweighting for estimation of fluctuation formulas”. In: *arXiv preprint arXiv:2504.11968* (2025).
- [4] R. Gastaldello, A. Henrot, and I. Lucardesi. “About the Blaschke-Santaló diagram of area, perimeter and moment of inertia”. In: *Journal of Convex Analysis* 31 (2024), pp. 563–602.

Oral presentations

- 12th French Biennial of Applied and Industrial Mathematics (SMAI2025), *Efficient computation of transport coefficients using the Green–Kubo formula with Girsanov’s reweighting*, Jun 2025
- “Sampling High-Dimensional Probability Measures with Applications in (Non)Equilibrium Molecular Dynamics and Statistics” CECAM Summer School at University of Birmingham, *Efficient computation of transport coefficients using the Green–Kubo formula with Girsanov’s reweighting*, Jul 2025
- CERMICS, ENPC, Paris, Young Researchers Seminar, *On the diffusive-mean field limit on weakly interacting kinetic diffusions*, Oct 2025.
- ANR SINEQ final conference, GSSI, L’Aquila, *On the diffusive–mean field limit of the kinetic interacting particle system*, Oct 2025

Teaching

2025–2026	Teaching Assistant , ENPC, Champs-sur-Marne, France. Course Analysis and PDEs, first-year engineering students, 18h.
2024–2025	Teaching Assistant , ENPC, Champs-sur-Marne, France. Course Analysis and PDEs, first-year engineering students, 36h.
2024–2025	Project coordinator , ENPC, Champs-sur-Marne, France. Supervision of 4 first-year engineering students, Subject: <i>Simulation of Hamiltonian dynamics and sampling of the underdamped Langevin dynamics</i> , 15h.
2023–2024	Project coordinator , ENPC, Champs-sur-Marne, France. Supervision of 4 first-year engineering students, Subject: <i>Introduction to the computation of transport coefficients in molecular dynamics.</i> , 15h.